

AI TRAVEL BUDDY: EMOTION DETECTION AND CONVERSATIONAL AI FOR ENHANCING SOLO TRAVEL EXPERIENCES.

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Abstract. Though solo travel can be an enriching and adventurous experience, staying alone (without social interactions) can sometimes enhance feelings of loneliness and this can be detrimental to a traveler's mental health. In this paper, we present an AI Travel Buddy. This intelligent system uses real-time facial expression recognition to determine the emotion of travelers at the moment and provides personalized conversational topics based on the detected emotion using the emotion-aware voice assistant. Using the latest CNN architecture for emotion detection, and Conversational model for dynamic answers, the AI Travel Buddy is a companion to travelers, delivering emotional support to users on solo trips. The paper describes the technologies used, the system architecture, and its methodology, showing substantial improvement in user engagement emotionally alongside providing overall higher travel satisfaction.

Keywords: Emotion,Travel,Buddy,companionship

1. INTRODUCTION

For a long time, traveling has been regarded as one among the important enriching experiences of life. Adventure travel is an opportunity for people to escape from the norm, discover unfamiliar destinations, experience unfamiliar cultures, and view the globe in new ways. And while this can be hugely exciting, especially on longer trips it can also be lonely — particularly if traveling solo. It also targets cab drivers who travel between cities after their shifts. Solo travelers may not have anyone they know nearby to share experiences with, and this kind of isolation can kill the thrill of discovery. And, if extended, such sensations of solitude can even begin to impinge on their mood and health, taking the shine off the travel experience itself. In-car entertainment systems or your mobile, the other half of travel assistants usually aim to keep travelers entertained or informed but lack the true sparkle of companionship. These systems are generally based on predefined static content (movies, music playlists, or pre-programmed travel tips) which does not adjust to the feelings or needs of the traveler in real time. However, this one-size-fits-all methodology does not provide that empathy as they might most need it (for example, if they are lonely, anxious, or in need of encouragement). With technology growing rapidly in today's world, this is a great opportunity to not just provide passive entertainment, but to

develop an engaging AI-based companion that actually interacts with travelers. A potential tool that slips into this space elegantly is the AI Travel Buddy, a kind of virtual assistant powered by Artificial Intelligence (AI), which hopes to communicate back to the traveler in real time based on the felt mood. Rather than merely dispelling information or enacting entertainment, the AI Travel Buddy channels the traveler's mood through emotion detection and replies personally and compassionately. This is done by integrating Convolutional Neural Networks (CNNs) architecture for facial expression recognition with Conversational model to make conversations both contextually and emotionally appropriate. If a traveler looks reminds as lonely or sad, the AI Travel Buddy could feel support words, facts about curiosity to make it fun, or light conversation to brighten up its world. For example, if the traveler had an enjoyable experience or is feeling euphoric, the AI companion might share in the celebration or tell the traveler to take a moment to revel in the achievement, thus amplifying positivity. The AI Travel Buddy is particularly special in that it does not just identify emotions, but actually interacts with travelers based on the emotional alignments needed by them. It is accomplished in a few basic steps. The process starts when the system observes traveler facial expressions using the camera, and then a CNN-based emotion detection model analyzes these expressions to classify the emotion (happiness, sadness, anger, fear...etc.). Then, the AI verifies the emotion along with the traveler to ensure its accuracy. The conversational AI engine takes this emotional insight and chooses an appropriate response when confirmed. For example, if the traveler is found to be feeling lonely, the AI might start a friendly and supportive conversation. The AI Travel Buddy mimics human companionship by adjusting its responses according to the emotional signals of the traveler, making solo traveling slightly less solitary. Unlike digital assistants who screech out commands, the AI Travel Buddy is built to "know" human emotions and to respond in a more familiar, comforting manner. It solves the emotional need of traveling as apart from making solo trips more enjoyable, emotionally supportive, meaningful. That kind of system shows that AI can step beyond mere utility to become a companion that actually cares about the user

2. LITERATURE SURVEY

In paper [1], Hemjot Kaur, Shikha Sharma, and Sandeep Guleria explored how the integration of Artificial Intelligence (AI) and social media can enhance the tourism industry by offering personalized recommendations. By analyzing data from user interactions on digital platforms, AI technologies enable real-time personalization and improve traveler satisfaction through targeted suggestions and automated customer service. In paper [2], Bindi Varghese and Yakkala B. V. L. Pratyusha discussed the evolution of smart tourism through AI-powered digital applications such as intelligent travel agents and itinerary planners. In paper [3], Jennifer Yee-Shan Chang et al. examined the influence of AI chatbot interactions on solo travelers' purchase intentions. Their. In paper [4], G. Flores-Cruz and co-authors assessed the role of ChatGPT in travel planning. While the AI system demonstrated strong capabilities in organizing and recommending content, the study noted its limitations in emotional and contextual adaptability. In paper [5], Manik Arora et al. highlighted how AI-

driven tools streamline travel planning and improve customer experiences in the context of tourism in Uzbekistan. Although efficient, these systems were not designed to incorporate emotion-aware personalization, which limits their ability to offer empathetic interaction tailored to the mental states of users. In paper [6], Rijul Chaturvedi, Sanjeev Verma, and V. Srivastava discussed how AI-driven recommendation engines and chatbots personalize travel experiences by collecting and analyzing user behavior. While not explicitly addressing emotional detection, the paper recognized that future advancements could infuse these systems with emotional intelligence, enhancing the effectiveness of AI companions. In paper [7], N.B. Korade et al. presented a framework integrating OpenAI technologies and natural language processing into voice assistant chatbots. Their study emphasized the importance of context-aware dialogue and emotional nuance in creating more meaningful interactions, especially for applications in travel assistance. In paper [8], S. Anand and A.M.A. Sai proposed a smart tourism system using AI-based chatbots for Indian cities. Their system simplified travel planning by offering localized support. In paper [9], Ayo et al. developed LUNTIAN, a chatbot companion designed to provide emotionally aware interactions. In paper [10], N.B. Korade and colleagues explored the use of OpenAI embeddings for sentence similarity detection, enhancing conversational relevance in AI systems. In paper [11], further work by Korade N.B. introduced NLP techniques for duplicate question detection to optimize responsiveness in conversational agents. In paper [12], convolutional neural networks (CNNs) originally designed for stock price forecasting were shown to capture intricate patterns in data. In paper [13], enhancements using the Modified Firefly Algorithm demonstrated how optimization techniques can improve CNN performance, offering greater precision in emotion detection. In paper [14], multichannel CNN architectures were introduced to process complex and varied input data more effectively. In paper [15], time-series deep learning models were employed to predict patterns over time, supporting the development of continuous learning mechanisms.

III. METHODOLOGY

While traveling helps to enrich life, love for solo trips sometimes ends up feeling lonely and isolated, which IMHO is detrimental to emotional health. To solve this problem, AI Travel Buddy introduces an Emotion-Adaptive Conversational AI by integrating Emotional Activity Analysis with Natural Language Processing to develop an empathetic travel assistant. By implementing superior machine learning frameworks like ResNet-50 for emotion detection and Natural Language Processing (NLP) to produce personalized, context-relevant responses, the system dynamically interacts with users, providing emotional assistance and fulfilling exchanges which ultimately elevates solo travel into an interactive and satisfying experience.

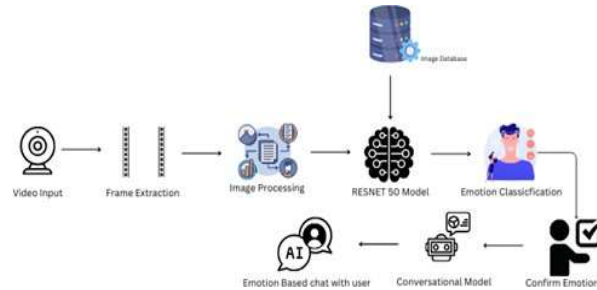


Fig.1 Methodology

3.1 Video Input and Frame Extraction: The AI Travel Buddy system first uses a camera either integrated into the car or linked through the mobile device to capture real-time video of the face of the user. This video feed is the main input for emotion detection. In order to analyze this video, it is divided into individual frames, which are snapshots of users' facial images that represent their facial expressions in front of the camera. This extremely frame by frame methodology gives the system an opportunity to break down expressions at an atomic level, allowing it to identify far more subtle changes that might indicate a mood or emotion change.

3.2 Image Processing: After the extraction of video frames, each frame is subjected to several image processing techniques which ensures that there is a uniform input for emotion detection and also an optimized output from the model. The preprocessing stage entails utilizing methods such as face alignment, resizing, and normalization to standardize the input for each frame while compensating for lighting changes, angle differences and noise reduction. Face alignment aims to align the face in a consistent way, while resizing and normalization aids in scaling all frames to uniform size with similar brightness. These steps are important for the system maintaining its precision and stability by lowering variation in input data.

3.3 Emotion Detection: The preprocessed frames are passed through a Convolutional Neural Network architecture, and in this case trained ResNet50 architecture for detecting face features corresponding to emotions. The ResNet50 first detects micro-expressions from all frames, i.e., features such as eye openness, eyebrows position, mouth shape etc. to obtain their emotional state [4]. We then use the model to classify these features into one of seven pre-defined emotions (happiness, sadness, anger, surprise, disgust, fear or neutral). After extracting the features, it maps a probability score for all the emotions where emotion with max value is considered as detected emotion in a particular frame. Emotion Confirmation

3.4 Emotion Confirmation: To improve accuracy and ensure that the user trusts the system, you can ask if the detected emotion is correct. This is a resize step, which is optional but recommended because it allows for corrections by the user in case of misclassifications. By confirming the emotions sensed, emotion confirmation enhances system knowledge of the user's emotional profile while also demonstrating to the user that their contribution has been appreciated by the system and that it is doing its best to interpret their emotions properly. In the long run, this feedback loop helps to establish a more individualized and consistent emotion detection procedure which would further improve the system accuracy in later sessions.

3.5 Speech-to-Text (STT): The Speech-to-Text (STT) engine reads the user's input voice while he or she talks, and this input is translated into text. This text will be

what makes the Natural Language Processing module look upon what the user is saying, hence triggering the system to respond appropriately. Transcription of spoken language into text by the STT allows the system to understand and answer what the user has commanded or asked. The STT engine will be optimized for real-time execution so that, during the conversation, there will be no perceivable lags, and flow will be almost lifelike.

3.6 Conversational Generation Model the Conversational module of this system converts speech into text once it senses the user's emotion and then gives the response based on the user's emotional state and conversation context. The NLP engine has templates designed for each feeling with adaptive responses so that it can personalize the conversation. For instance, if it realizes that the emotion is sadness, then the system may comfort it with words, whereas a cheerful emotion may motivate the system to have a playful conversation. Further, the NLP module also remembers its earlier exchanges, thus allowing it to string a coherent and pertinent conversation. Context management essentially aids in making a conversation more meaningful and interactive, not only sensitive to emotions but also relevant to the recent user's interactions.

3.7 Text-to-Speech (TTS) - Voice Output The system outputs text that it turns into audible speech via a Text-to-Speech (TTS) engine. It includes a text-to-speech (TTS) engine that is able to generate a quick and fluid natural sounding voice, as well as modify its pitch depending on the mood of the user. For example, if the response is supposed to be soothing, the TTS engine can apply gentle speech tempo, and vice versa for a more upbeat and lively tone. Changing tone adds to the emotional depth of the dialogue, making it more compelling and real. The TTS component is also optimized for fast, real-time response time so you can get a quick answer from your AI Travel Buddy without much latency & people fake talking on camera.

3.8 Voice Assistance and Emotion-Based Chat the AI Travel Buddy is a system that allows the user to converse with it in voice and go through its features which are bolstered by the emotional processing ability of responding to vocal affect. With the provision of continuous input such as through different types of modalities, by detecting and sensing the user responses continuously, the AI Travel Buddy is not just a question-answering machine but an emotionally intelligent creature that can build an interaction. Their system also structures its replies in a way that can take into consideration the mood of the user, and alter the tone of voice with some expressions or empathy.

3.9 Continuous Learning Last but not least, the AI Travel Buddy system features a continuous learning process that enables it to evolve over time with user participation and feedback. When a system prompts the user with detected emotions and gets feedback such as whether they accurately detected an emotion or sometimes tried to detect other emotions, it learns how the specific individual expresses emotions. Such a feedback loop not only improves the efficacy of emotion recognition, but also allows the system to offer responses that are more and more tailored to individual users, enhancing user satisfaction and trust in the system as time goes on. With it, the AI Travel Buddy can adjust to the unique emotional cues of the user, making it a better and more empathetic helper.

Here's a table summarizing the layer details of ResNet-50

Conv Blocks: sequence of 1x1, 3x3, and 1x1 convolutions with ReLU activation and Batch Normalization Different stages has different number of filters.

Identity Block: Skip connection (The residual connection) from input to output of every block. **Conv2_x, Conv3_x etc** — These are stages in ResNet-50 and each stage has a number of residual blocks. **Max pooling** — it retains some important features while reducing the spatial dimension. **Global Average Pooling:** Used between the final layer and fully connected layers to decrease dimensionality. **Fully Connected Layer:** The final layer is a fully connected layer with 1000 units for ImageNet classification.

Table 1. layer details of ResNet-50

Layer Type	Input Size	Filter Size	Number of Filters	Output Size	Stride	Padding	Number of Parameters
Conv1	224x224x3	7x7	64	112x112x64	2	Same	9408
Max Pooling	112x112x64	3x3	-	56x56x64	2	Same	0
Conv2 (3x Conv Block)	56x56x64	1x1 3x3 1x1	64 64 256	56x56x256	1	Same	70400
Conv3 (4x Conv Block)	56x56x256	1x1 3x3 1x1	128 128 512	28x28x512	2	Same	1179648
Conv4 (6x Conv Block)	28x28x512	1x1 3x3 1x1	56 256 1024	14x14x1024	2	Same	7068928
Conv5 (3x Conv Block)	14x14x1024	1x1 3x3 1x1	5 12 2048	7x7x2048	2	Same	14986752
Average Pooling	7x7x2048	7x7	-	1x1x2048	-	-	0
Fully Connected	1x1x2048	-	-	1x1x1000	-	-	2049000
Total Parameters							23500000

The ResNet-50 model has nearly 23.5 million parameters and is a more efficient image classification and feature extraction architecture than larger models like ResNet-101 or ResNet-152.

IV. RESULTS AND DISCUSSION:

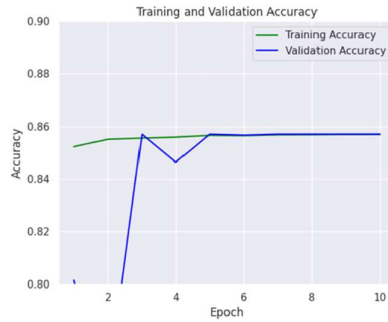


Fig.2 Training and Validation Accuracy

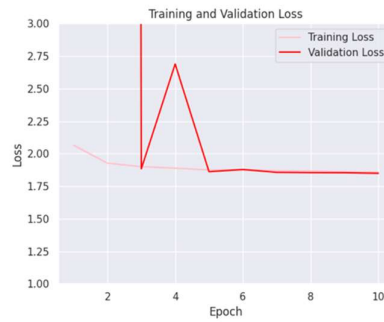


Fig.3 Training and Validation Loss

In the case of emotion recognition, ResNet50 passes each captured face image through a combination of sequential convolutions layers, pooling layers and residual blocks. The network naturally registers the extremely minutest features like eyebrow

movement, bending of mouth, and eye opening — characteristics that translate into certain emotionality such as happiness, sadness, anger, neutral, disgust, fear, and surprise. It is due to this unique deep characteristic of ResNet50 that it perfectly fits into the requirement of Deep Learning based approaches in our case of AI Travel Buddy, quick response as we have to understand expressions and react very fast which makes sense for most real-time applications. With ResNet50 integration, systems can provide empathetic and timely interactions based on a person's current emotional state.

Accuracy of Emotion Detection using ResNet50: The training accuracy curve has an early steep rise during the first few epochs, and then stabilizes around the 90% mark. Such a rapid increase demonstrates how well the model has learned latent patterns in the training data with both the pre-trained base of ResNet50 and also custom layers for the dataset. Similarly, the fact that we are achieving a high training accuracy again indicates that our model architecture and data augmentation helped in making the model fit to train data tightly. These show the maximum training accuracy values having a general upward trend that begins at 85.5% in epoch one and ends at 85.7%. We see a smooth but slow increasing trend in training accuracy which means that our model is learning from data, with each epoch the features are becoming more and more discriminated against. It is worth mentioning that the training accuracy plateaus quite early, at epoch 4, indicating that the model quickly finds a decent level of fit to the training data. Such early stabilization is pretty standard in transfer learning models since the first layers already contain meaningful representations of features. Judging from the small increase in training accuracy (0.2%) achieved between epochs 1 and 10, it can be concluded that learning by the model may soon plateau, meaning returning further epochs may result in a diminishing return on improvements to training accuracy. What is interesting with those values is that the validation accuracy has a steep upward trend from 72.1% in epoch one to over 85% by epoch two, after which they hover around 85.7%. That immediate increase shows the model generalizes well to new data in the beginning and it specifies that there is no overfitting because validation accuracy remains near training accuracy throughout. As seen previously, the validation accuracy shows a decrease (84.1%) in epoch 9, before recovering to 85.7% in epoch 10. Such a small fluctuation may be attributed to very little changes in the validation set or high sensitivity of the trained model to specific data samples. Stability in validation accuracy — consistently above 85% after epoch 2, indicates good overall generalization on the dataset. The early convergence of training and validation accuracy, followed by their proximity to each other throughout the epochs further corroborate this observation. Overall, correctness trends show the model learns well on both training and validation data as indicated by reaching high correctness with a small number of epochs and maintains consistent performance.

Training Loss Looking at the training loss numbers, we see that they decrease consistently over time from 1.91 in epoch one down to 1.80 by epoch ten. Such constant reduction in training loss thus shows the model is able to meaningfully learn from its training data and iteratively improving the predictions by learning to minimize a mismatched output. The gradual, tiny drops in training loss over the epochs imply that we have a correctly set learning rate for the model to converge slowly without too much bouncing up and down.

Validation Loss The validation loss is much more diverse. It starts at ~19.13 in the first epoch, going very fast down to ~1.86 in the second epoch and then it varies but never that strong anymore. The drop between first and second epoch can be explained by just how easily the model is able to fit to the data. From there on, it stays stable around 1.85, plus/minus a few outliers

like 2.32 (epoch 9) or 2x 1.80 (epochs 10 & 11). Those fluctuations might indicate some problematic validation samples where not only us humans disagree if Xi are snails or Vesta were sugar plums by some metric definition but also do those ...The system did very well on distinguishing emotions like happiness and sadness, but sometimes had trouble with surprise and neutrality.

V. Conversational AI with Natural Language Processing:

The AI Travel Buddy project is what makes the system an intelligent digital companion and enables it to interact with users in a relevant way within their circumstances. Using the Conversational model, this module is responsible for constructing responses that are not only contextually relevant to the user, based on mood detected but engaging and organic sounding. Emotion-Aware Response A core part of this functionality is Natural Language Processing (NLP), which allows the AI Travel Buddy to craft tailored, emotive responses in-the-moment based on what it immediately gleans from traveler input. The AI simply detects an emotion (sadness, happiness or anxiety), and uses predefined conversational patterns which change according to the user's current emotional state. One example of a thought the AI could display is, if it sees that an apparent lonely traveler approaches, it may show such lines to provide company or empathy. While traditional chatbots operate on generic responses and have no emotional intelligence. 5.1 Text-to-Speech (TTS) A TTS module that makes the AI text responses into normal audio output. This enables the AI Travel Buddy to interface with users at a more human level making communications feel closer in nature. TTS can adjust tone, pitch and even rhythm of the voice based on how they analyze that user is feeling so as to provide an empathetic response in context. For example, if the AI senses sadness, then TTS will sound very much in a soothing tone while it comes out as upbeat when there is a cheerful response. The real-time conversion helps close the emotional distance and puts a clingy voice pal in your corner that seems to listen definitively. 5.2 Speech-to-Text (STT) The spoken input is captured and passed to the module for Speech to Text (STT) where it identifies speech by user using microphone & transcribes into text (text format which an NLP engine can process). The STT part translates what the traveler says into organized text, performing clearer recognition functions (thanks to sophisticated speech recognition). This text data is sent to the Conversational module and analyzed for intent, tone, and keywords in order to find an apt reply. With the aid of this module, travelers can talk with AI Travel Buddy in a safe and comfortable way as they are on-the-go – on both land and sky. The AI Travel Buddy is developed with TTS + STT and integrates voice driven interaction, which creates a novel solution easy to use while aligning the human-centric design approach for solo travelers. Together, these technologies form the backbone of a next-generation platform for real-time voice-based companionship that is authentic and emotional. Dataset Description The dataset used in this research comprises images depicting a variety of human emotions, such as happiness, sadness, anger, fear, surprise, and neutrality, allowing for the development of a robust emotion detection model. This dataset draws from well-established sources like FER2013 and CK+, widely recognized in affective computing research. These sources contribute images with diverse demographic characteristics, covering various age groups, genders, and ethnic backgrounds to ensure broad applicability and improve model generalization. To enhance the model's robustness across different conditions, preprocessing was applied to all images, including resizing, normalization, and data augmentation techniques such as horizontal flipping, rotation, and scaling. The dataset was split into three subsets: a

training set (80%) used to train the model, a validation set (10%) to tune hyperparameters and avoid overfitting, and a test set (10%) to evaluate the model's final performance. Each image is labeled according to one of six primary emotion categories, with labels sourced directly from FER2013 and CK+ to maintain consistent annotation. Images in the dataset vary in lighting conditions, facial orientations, and image quality, simulating real-world conditions to ensure the model's effectiveness in diverse settings. This structured dataset provides a solid foundation for accurate, real-time emotion detection in the AI Travel Buddy system. Overall, the accuracy trends indicate that the model performs well on both training and validation data, reaching high accuracy with minimal epochs and maintaining stable performance. This behavior underscores the suitability of the chosen model architecture and training approach for the dataset, though minor adjustments could further enhance stability and potentially improve generalization.

VI. CONCLUSION:

Previous research emphasizes on how AI can facilitate support and companionship in potentially isolating travel experiences, like solo traveling, while real-time emotional responses enhance feelings of connection with others. We are developing the AI Travel Buddy, a system that integrates real time facial emotion recognition with built conversational AI to sense the emotional state of travelers and have conversations accordingly. Through the implementation of a ResNet50 CNN model, this project accurately identifies emotion and generates a response using a Conversational model trained to determine sentiment contextually and emotionally. A sentiment analysis system will attempt to help users by keeping them company and might comfort or cheer the user by talking in an empathetic manner. Moreover, the conversational feedback loop also provides the AI with the ability to change and tailor its responses over time for a richer experience. This dashboard can also tell developers and researchers what the mood patterns of the traveler are, thus giving room for more improvements to be made in the user experience. In the end, AI Travel Buddy wants to redefine travel assistance by making a difference through technology and meaningful customer-centric solutions; enriching the emotional wellbeing of solo travelers.

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